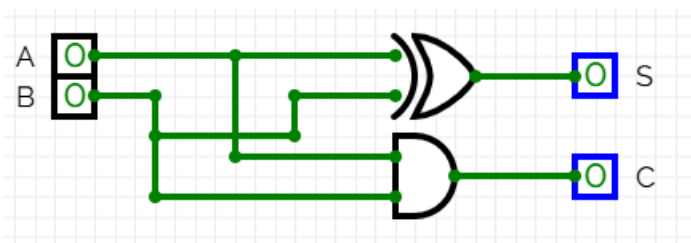
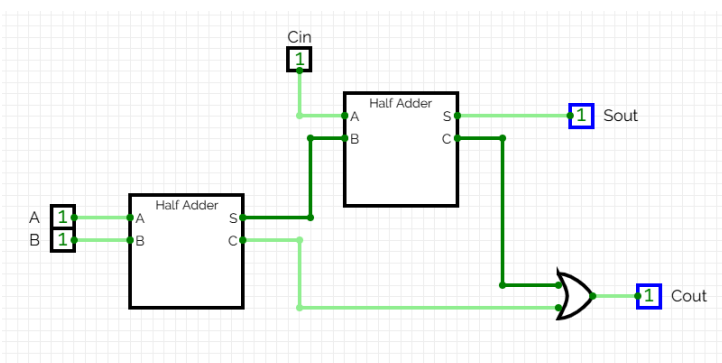
Module 3 Assessment – CircuitVerse ALU

Function F(A,B)	F2	F1	F0
A+B	0	0	0
A-B	0	0	1
A+1	0	1	0
A-1	0	1	1
A AND B	1	0	0
A OR B	1	0	1
A XOR B	1	1	0
NOT A	1	1	1

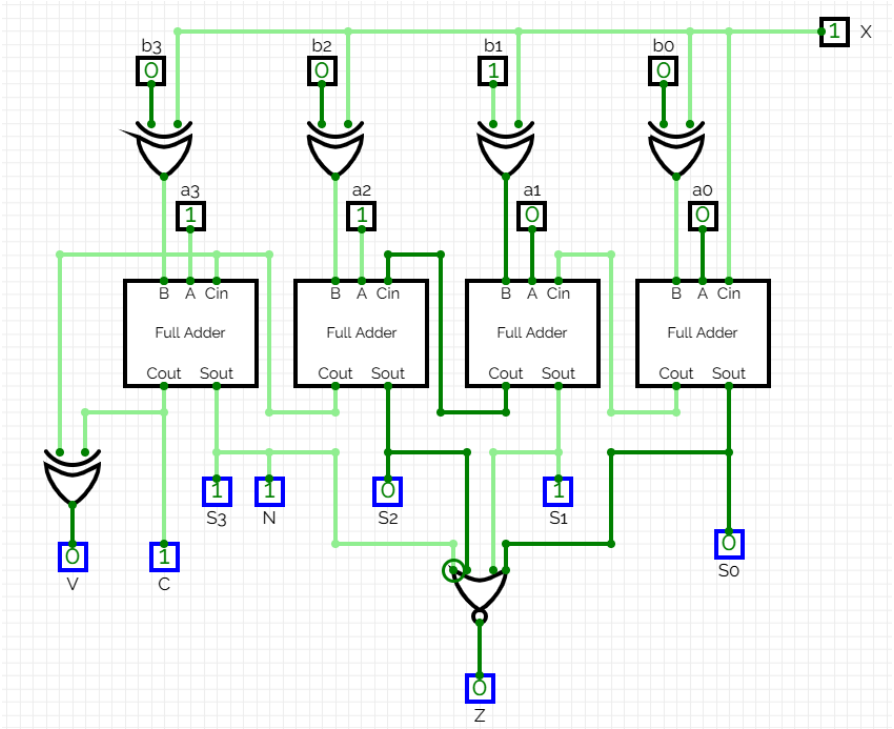
HA Screenshot:



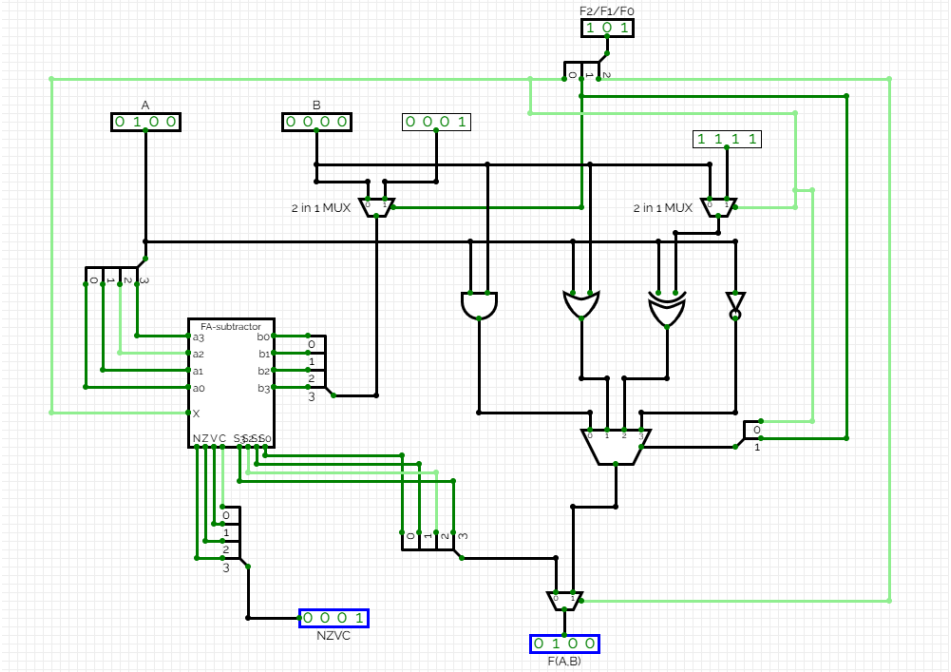
FA Screenshot:



FA-Subtractor Screenshot:

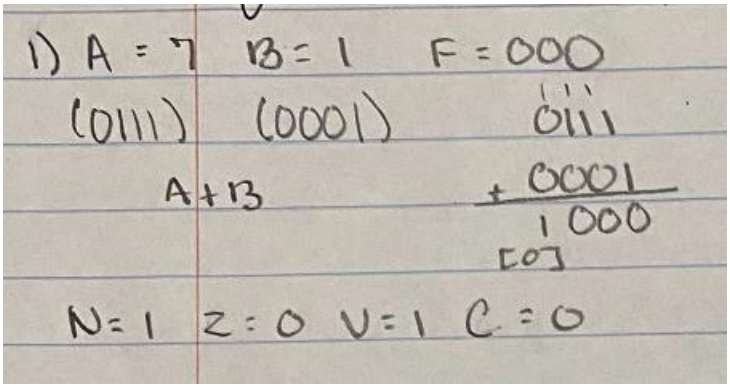


ALU Subcircuit Screenshot:

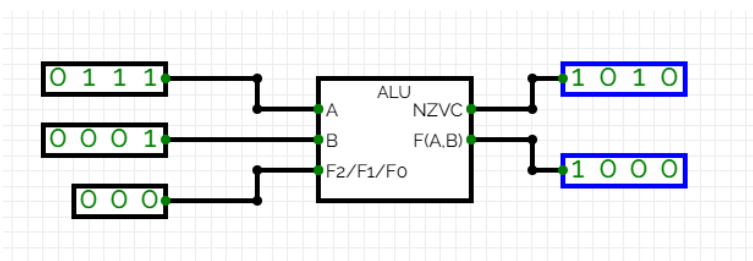


1) Test case: A = 7; B = 1; F = 000

Prediction:

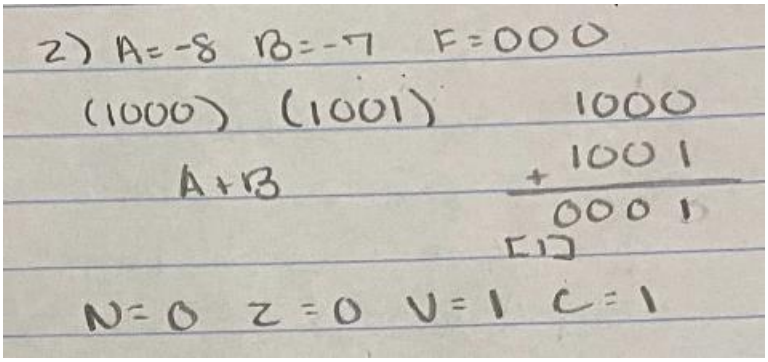


Main Circuit:

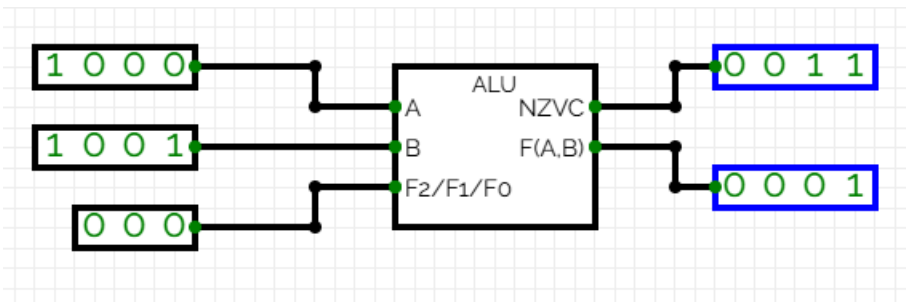


2) Test case: A = -3; B = -7; F = 000

Prediction:

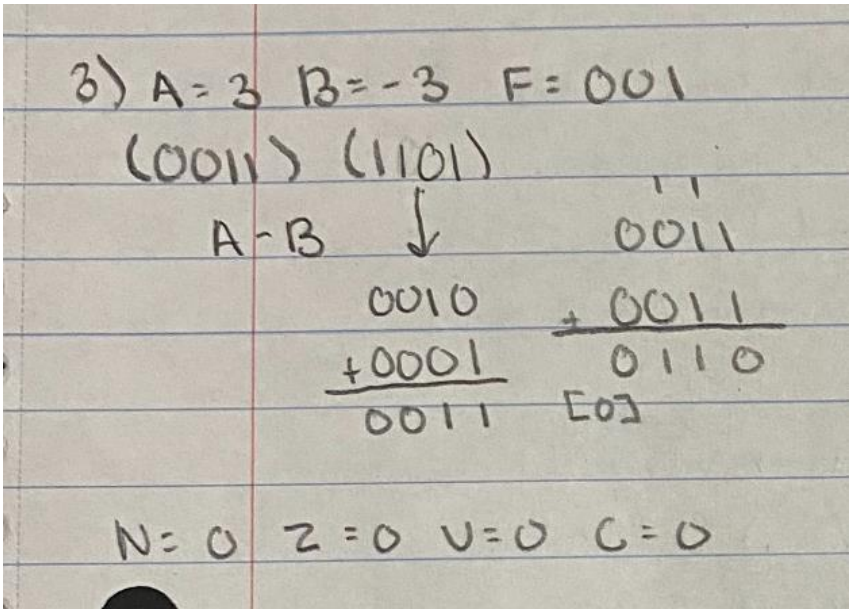


Main Circuit:

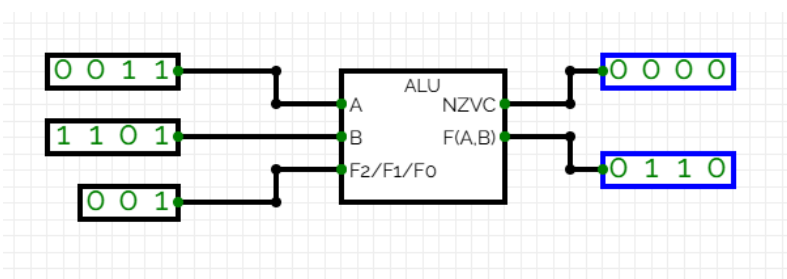


3) Test case: A = 3; B = -3; F = 001

Prediction:

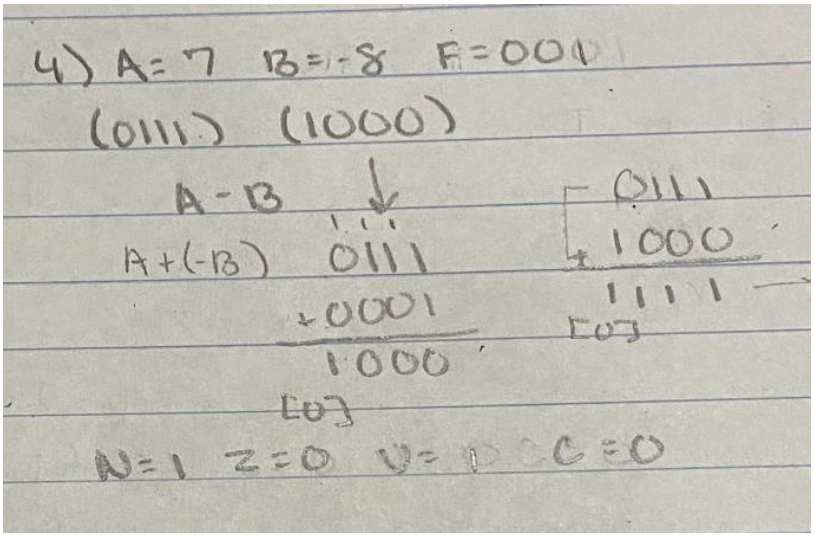


Main Circuit:

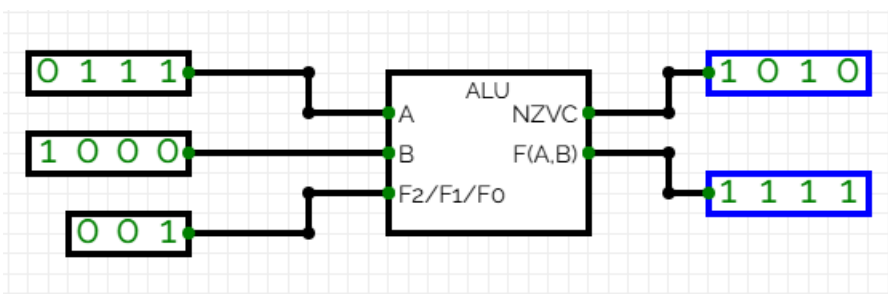


4) Test case: A = 7; B = -8; F = 0001

Prediction:

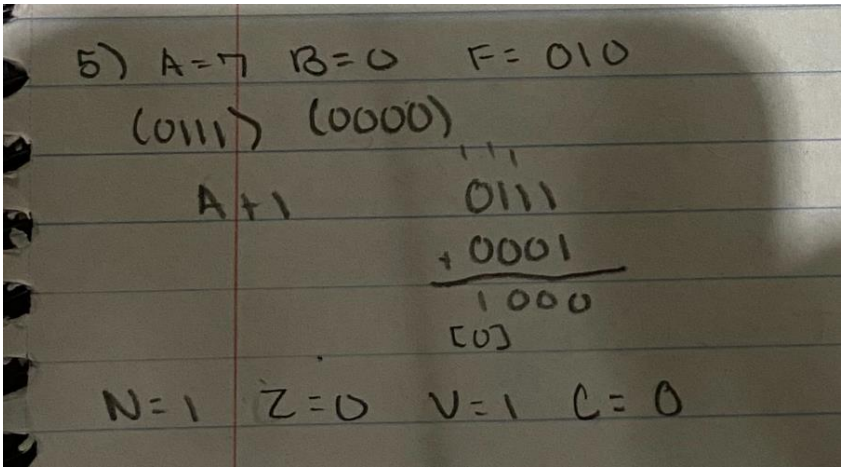


Main Circuit:

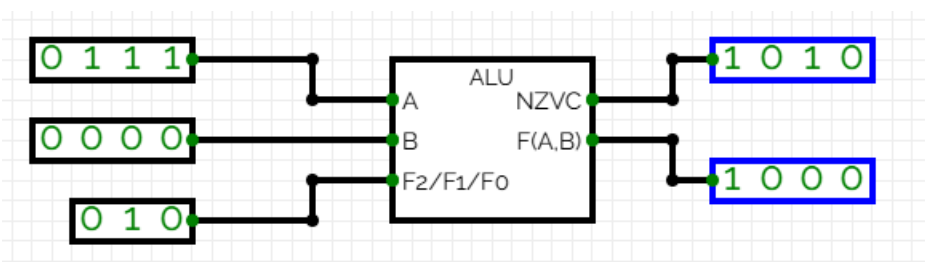


5) Test case: A = 7; B = 0; F = 010

Prediction:

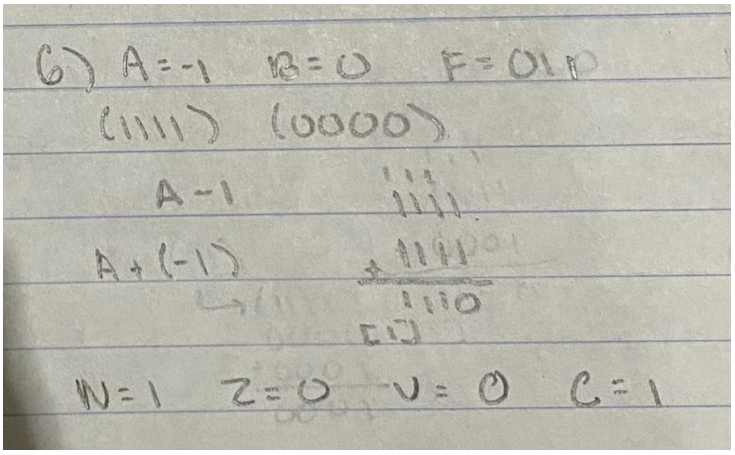


Main Circuit:

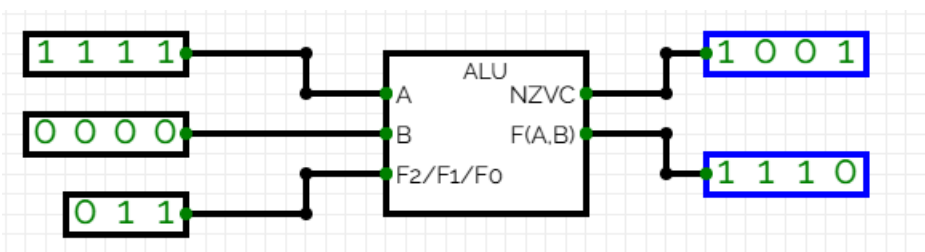


6) Test case: A = -1; B =0; F = 011

Prediction:

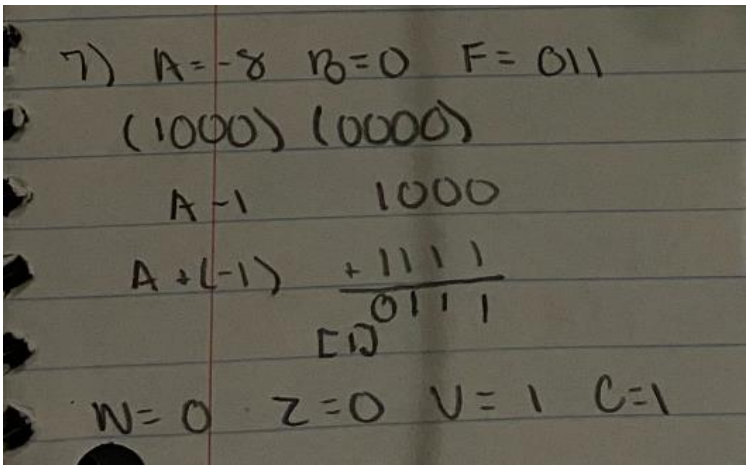


Main Circuit:

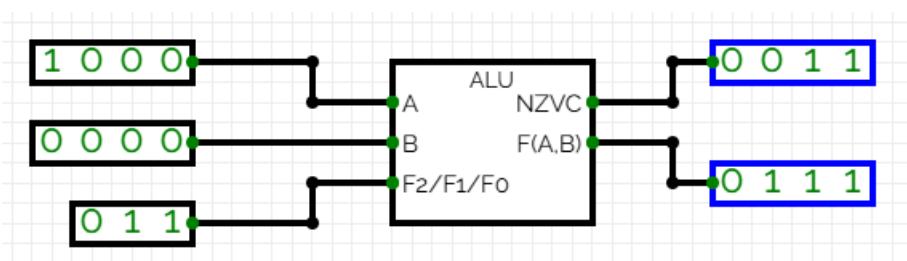


7) Test case: A = -8; B = 0; F = 011

Prediction:

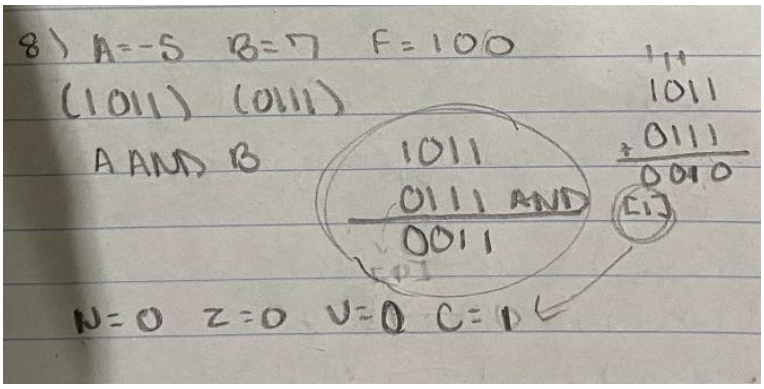


Main Circuit:

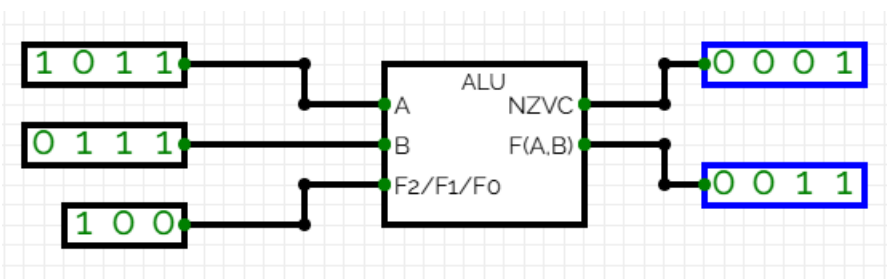


8) Test case: A = -5; B = 7; F = 100

Prediction:

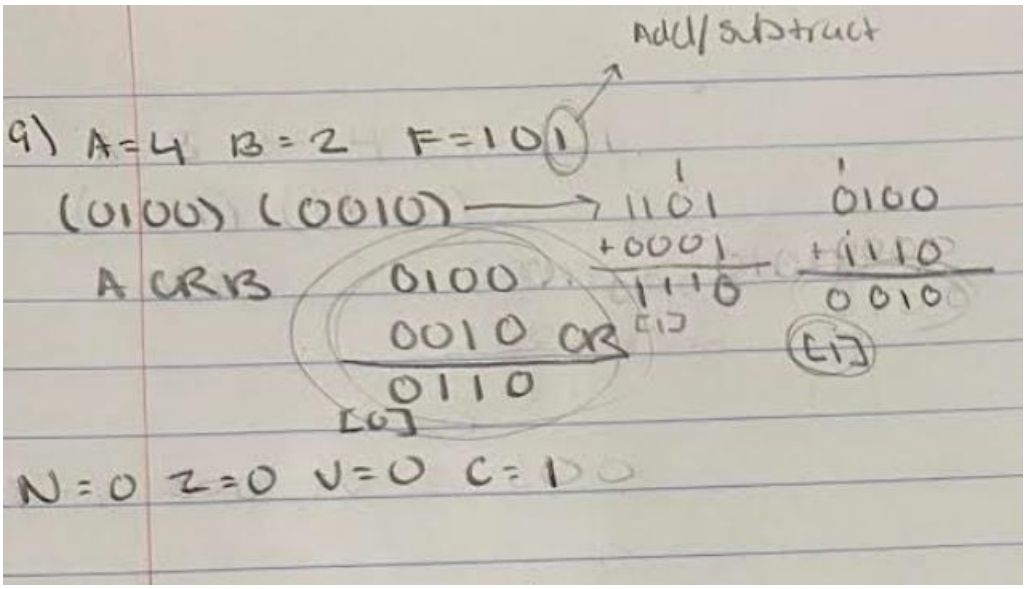


Main Circuit:

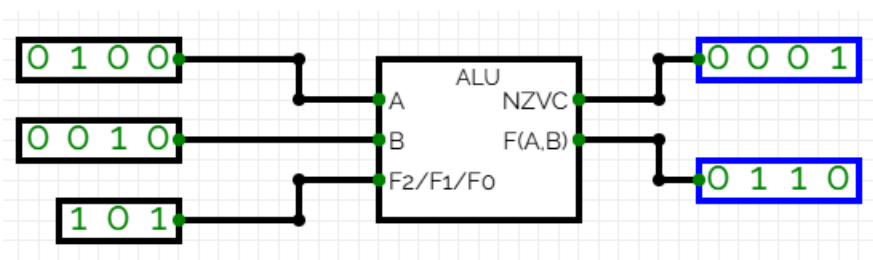


9) Test case: A = 4; B = 2; F = 101

Prediction:

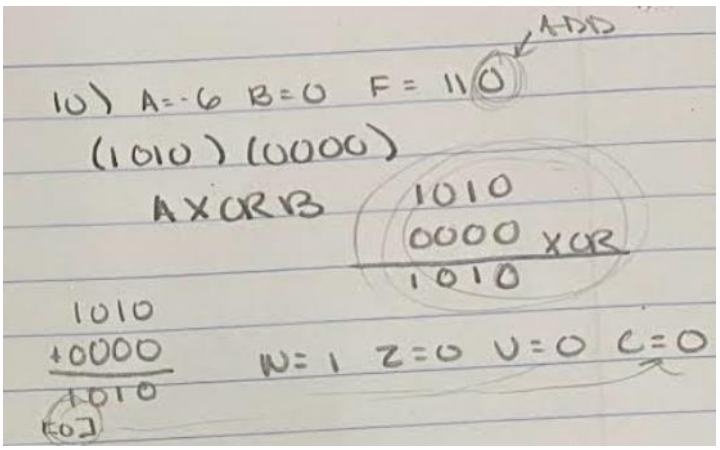


Main Circuit:

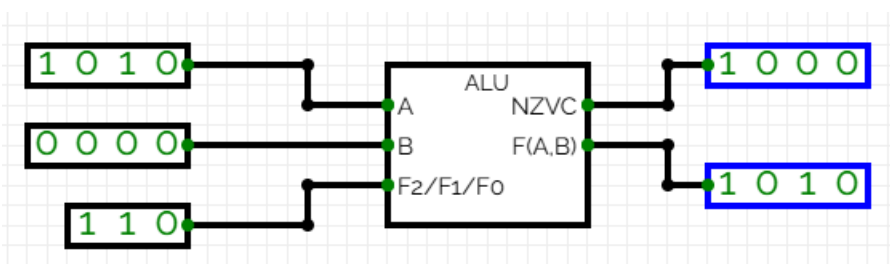


10) Test case: A = -6; B = 0; F = 110

Prediction:

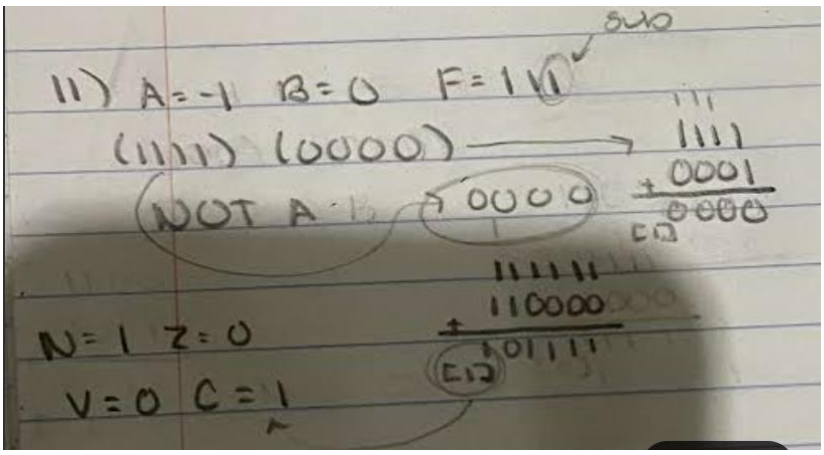


Main Circuit:

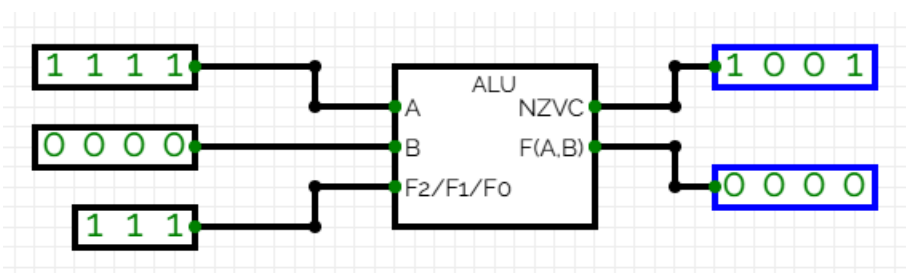


11) Test case: A = -1; B = 0; F = 111

Prediction:

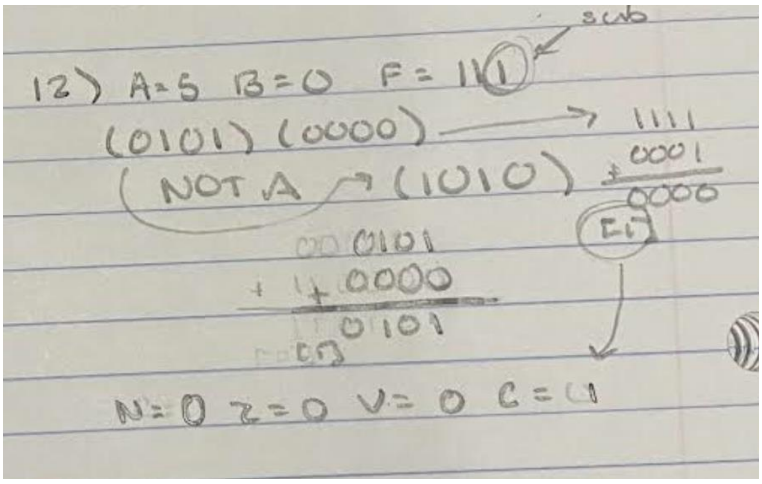
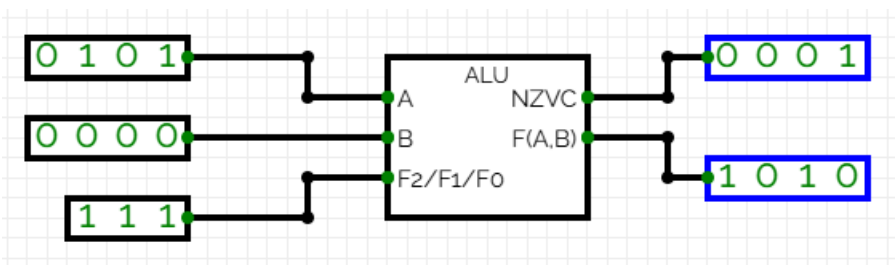


Main Circuit:



12) Test case: A = 5; B = 0; F = 111

Prediction:



Main Circuit: